

KEY R FUNCTIONS USED

- `plot()` – Creates line and scatter plots.
- `axis()` – Adds custom month labels to the x-axis.
- `barplot()` – Creates bar charts.
- `fluidPage()` – Creates the Shiny user interface.
- `selectInput()` – Provides chart selection.
- `renderPlot()` – Displays plots dynamically.
- `shinyApp()` – Runs the Shiny application.

SET 2 — CUSTOMER FEEDBACK ANALYSIS

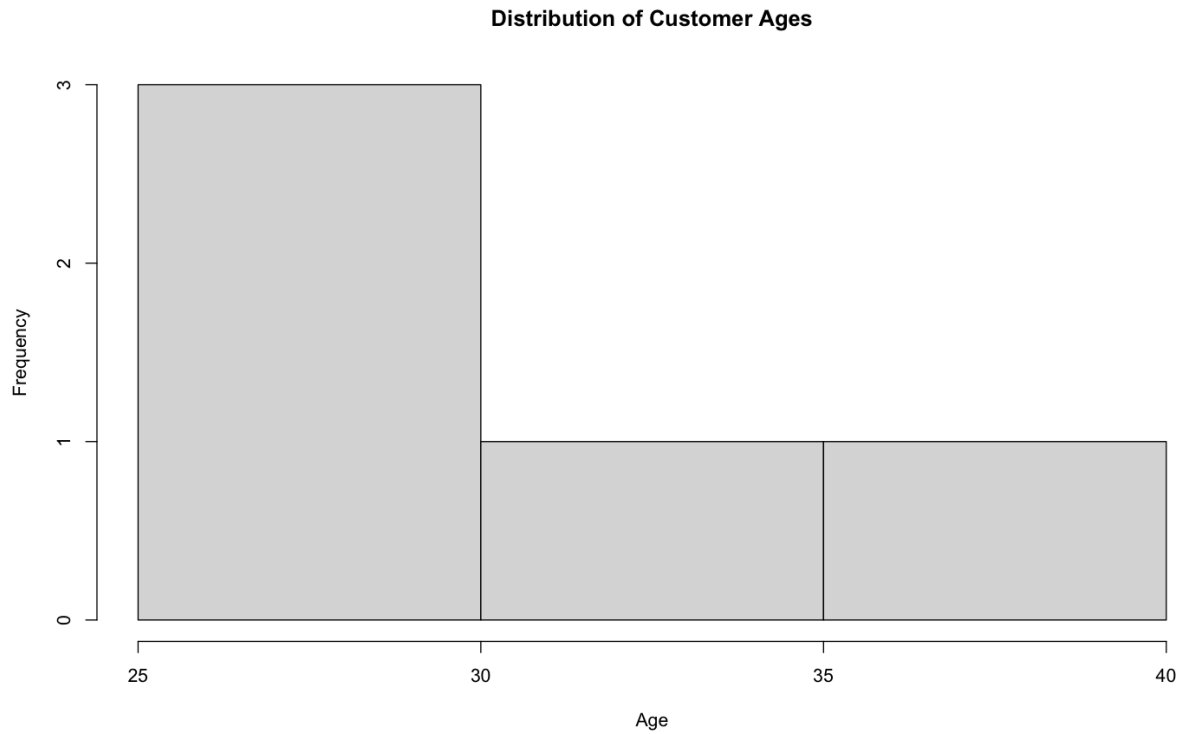
Dataset

Customer ID	Age	Satisfaction Score
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

PROGRAM 1 — Histogram

```
# Dataset
age <- c(25, 30, 35, 28, 40)
# Histogram
hist(age,
      main = "Distribution of Customer Ages",
      xlab = "Age",
      ylab = "Frequency")
```

Output: Histogram showing the distribution of customer ages.

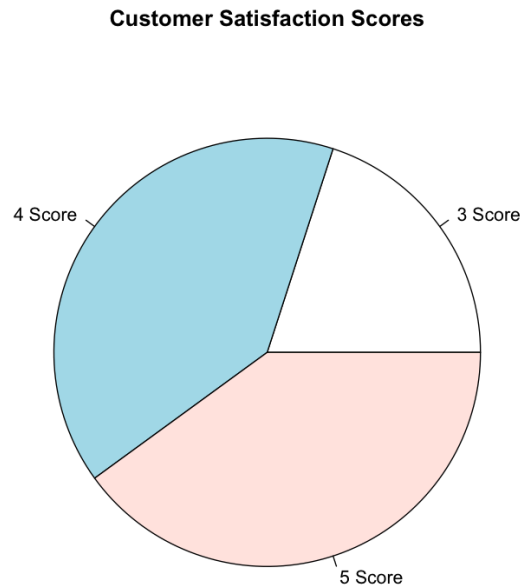


Observation: The customers have ages ranging from 25 to 40 years.

PROGRAM 2 — Pie Chart

```
# Dataset
satisfaction <- c(4, 5, 3, 4, 5)
# Count satisfaction scores
score_count <- table(satisfaction)
# Pie Chart
pie(score_count,
    labels = paste(names(score_count), "Score"),
    main = "Customer Satisfaction Scores")
```

Output: Pie chart representing the frequency of each customer satisfaction score.



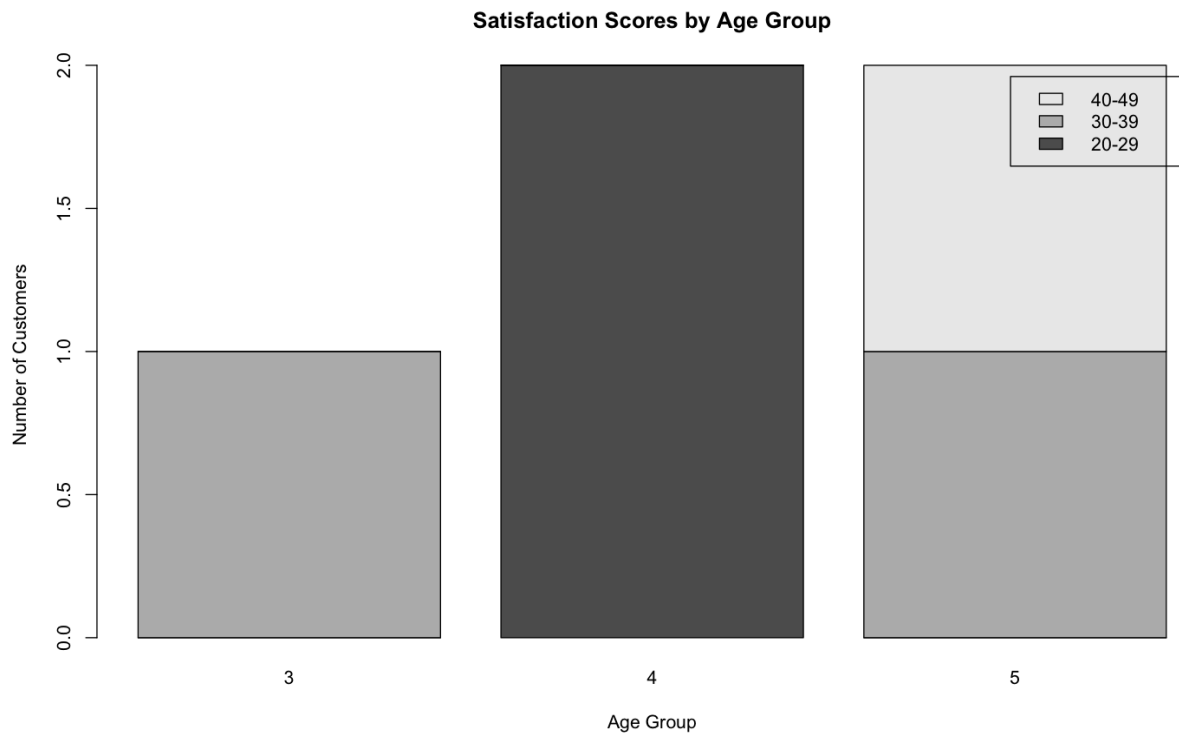
Observation: Satisfaction scores 4 and 5 occur twice each, while score 3 occurs once.

PROGRAM 3 — Stacked Bar Chart

```
# Dataset
age <- c(25, 30, 35, 28, 40)
satisfaction <- c(4, 5, 3, 4, 5)
# Create age groups
age_group <- cut(age,
  breaks = c(20, 29, 39, 49),
  labels = c("20-29", "30-39", "40-49"))
# Create table
data <- table(age_group, satisfaction)
# Stacked Bar Chart
barplot(data,
  beside = FALSE,
  main = "Satisfaction Scores by Age Group",
  xlab = "Age Group",
  ylab = "Number of Customers",
```

```
legend.text = TRUE)
```

Output: Stacked bar chart showing satisfaction scores across different age groups.



Observation: The chart displays the distribution of satisfaction scores for the 20–29, 30–39, and 40–49 age groups.

PROGRAM 4 — Word Cloud

```
# Install packages if required
install.packages("wordcloud")
install.packages("RColorBrewer")
# Load libraries
library(wordcloud)
library(RColorBrewer)
# Customer feedback
feedback <- c(
  "good service",
  "excellent service",
  "good experience",
  "friendly staff",
  "excellent service"
```

```
)  
# Convert feedback into words  
words <- unlist(strsplit(tolower(feedback), " "))  
# Count words  
word_count <- table(words)  
# Word Cloud  
wordcloud(names(word_count),  
          freq = word_count,  
          min.freq = 1,  
          scale = c(3, 0.5),  
          colors = brewer.pal(8, "Dark2"),  
          random.order = FALSE)
```

Output: Word cloud representing frequently used words in customer feedback.



A word cloud visualization of customer feedback. The words are arranged in a cluster, with 'service' being the largest and most central word. Other prominent words include 'excellent', 'good', 'friendly', 'staff', 'experience', and 'staff'. The colors used are shades of blue, orange, and grey.

experience
excellent
service
good friendly
staff

Observation: Words such as **service, excellent, good, friendly, and experience** are highlighted based on their frequency.

PROCEDURE

1. Create the customer age and satisfaction score vectors.
2. Use `hist()` to visualize the distribution of customer ages.
3. Use `table()` and `pie()` to represent satisfaction scores.
4. Divide customers into age groups using `cut()`.
5. Use `barplot()` to create the stacked bar chart.
6. Load the `wordcloud` and `RColorBrewer` packages.
7. Split the feedback into individual words and count their frequency.
8. Generate the word cloud using the word frequencies.

RESULT

Customer feedback data was successfully visualized using a **histogram, pie chart, stacked bar chart, and word cloud**.

KEY R FUNCTIONS USED

- `hist()` – Creates a histogram.
- `table()` – Counts frequencies.
- `pie()` – Creates a pie chart.
- `cut()` – Creates age groups.
- `barplot()` – Creates a bar chart.
- `strsplit()` – Splits text into words.
- `wordcloud()` – Creates a word cloud.
- `brewer.pal()` – Provides color palettes.

SET 3 — EMPLOYEE PERFORMANCE EVALUATION

Dataset

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92